Indian Statistical Institute

Final Examination 2016-2017

Analysis II, B.Math First Year

Time : 3 Hours   Date : 24.04.2017   Maximum Marks : 100   Instructor : Jaydeb Sarkar

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(i) Answer all questions. (ii) $B_r(a) = \{x \in X : d(x,a) < r\}$. (iii) $C[0,1]$ is the set of continuous real valued functions on $[0,1]$ with sup metric.

Q1. (15 marks) Determine the nature of the critical points of $f(x,y) = x^3 + 6xy + 3y^2 - 9x$.

Q2. (10 + 5 marks) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) Compute the partial derivatives of f at $(0,0)$. (b) Prove that f is not differentiable at $(0,0)$.

Q3. (15 marks) Let U be an open subset of $\mathbb{R}^n$, and let $f : U \to \mathbb{R}$ be a differentiable function. Let $a, b \in U$ such that U contains the line segment L from a to b (that is, $L = \{(1-t)a + tb : t \in [0,1]\} \subseteq U$). Prove that there is some $c \in L$ such that $f(b) - f(a) = f'(c)(b-a)$.

Q4. (10 marks) Let $a, h \in \mathbb{R}^n$, $a + h \in B_r(a)$, and let $f : B_r(a) \to \mathbb{R}$ be a C^2 function. Define the real-valued function η on $[0,1]$ by $\eta(t) = f(a+th)$, $t \in [0,1]$. Compute the second order derivative of η.

Q5. (10 marks) Let X be a metric space. Prove that the closure of a connected subspace of X is also connected.

Q6. (10 marks) Prove that a subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.

Q7. (10 marks) Let X be a compact metric space and $f : X \to X$ an isometry. Prove that $f(X) = X$.

Q8. (7 + 8 marks) (a) Let $t \in [0,1]$. Prove that the evaluation map $ev_t : C[0,1] \to \mathbb{R}$ defined by $ev_t(f) = f(t)$, $f \in C[0,1]$, is continuous. (b) Let K be a closed subset of $\mathbb{R}$ and let $S_K := \{f \in C[0,1] : f([0,1]) \subseteq K\}$. Use part (a) to prove that S_K is a closed subset of $C[0,1]$.

Q9. (8 + 7 marks) Let X be a metric space. Prove the following:

(a) The union of two intersecting connected subspaces of X is connected.

(b) The union of two compact subspaces of X is compact.